



Randy Crawford
Manager
Regulatory Assurance
225-381-4177

RBG-48275

10 CFR 50.73

February 8, 2024

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2023-006-00, Automatic Reactor Scram
Due to a Turbine Trip Following a Disconnect Failure at Switchyard

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report.

This document contains no commitments.

Should you have any questions, please contact Mr. Randy Crawford, Regulatory Assurance
Manager, at 225-381-4177.

Respectfully,

A handwritten signature in black ink, appearing to read 'Randy Crawford', with a long, sweeping horizontal line extending to the right.

Randy Crawford

RC/db

Enclosure: Licensee Event Report 50-458 / 2023-006-00, Automatic Reactor Scram Due to
a Turbine Trip Following a Disconnect Failure at Switchyard

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station

Enclosure

RBG-48275

**Licensee Event Report 50-458 / 2023-006-00, Automatic Reactor Scram Due to a Turbine
Trip Following a Disconnect Failure at Switchyard**



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

River Bend Station, Unit 1

☒ 050
☐ 052

2. Docket Number

458

3. Page

1 OF 3

4. Title

Automatic Reactor Scram Due to a Turbine Trip Following a Disconnect Failure at Switchyard

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
12	13	2023	2023	- 006 -	00	02	08	2024	N/A	☐ 052	N/A

9. Operating Mode

Mode 1

10. Power Level

100%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Randy Crawford, Regulatory Assurance Manager

Phone Number (Include area code)

225-381-4177

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
A	FK	DISC	N/A	Y	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year
N/A	N/A	N/A

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced type written lines)

On December 13, 2023, at 01:02 CST, with River Bend Station, Unit 1, operating in Mode 1 at 100% power, a turbine trip signal resulted in an automatic reactor scram. All control rods fully inserted and there were no complications. All systems responded as designed.

The turbine trip signal was caused by a switchyard disconnect that was not properly seated in the saddle of the disconnect fingers, which resulted in an open circuit condition on the output of the main generator.

Following a trip of the Reactor Feedwater System due to a Reactor Feedwater Heating isolation, Reactor Core Isolation Cooling was utilized to maintain reactor vessel inventory. The Reactor Feedwater System was restarted once the Reactor Feedwater Heating isolation signal was reset. Reactor water level control was transitioned back to the Reactor Feedwater System.

This report is made pursuant to 10 CFR 50.73(a)(2)(iv)(A), any event or condition that resulted in manual or automatic actuation of the Reactor Protection System.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME River Bend Station, Unit 1	☒ 050	2. DOCKET NUMBER 458	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER 006	REV NO. 00

NARRATIVE**EVENT DESCRIPTION**

On December 13, 2023, at 01:02 CST, River Bend Station (RBS), Unit 1, was operating in Mode 1 at 100% power when a turbine trip signal resulted in an automatic reactor scram. All control rods fully inserted and there were no complications. All systems responded as designed.

Following the scram, the Reactor Feedwater System (FWS) [SJ] tripped on low suction pressure caused by isolations to both trains of Reactor Feedwater Heating [SJ]. At 01:08 CST, reactor water level control was transitioned to the Reactor Core Isolation Cooling (RCIC) [BN] system to maintain inventory of the reactor vessel. At 01:10 CST, Reactor Feedwater System Pump B (FWS-P1B) [SJ] was restarted. At 01:14 CST, reactor water level control was transitioned back to FWS and RCIC was secured. Reactor pressure was maintained by the Turbine Bypass Valves (JI).

This event was reported under 10 CFR 50.72(b)(2)(iv)(B), as any event or condition that results in actuation of the Reactor Protection System (RPS) when the reactor is critical and 10 CFR 50.72(b)(3)(iv)(A) Specified System Actuation as result of Group 3 isolations (EN 56887).

This report is made pursuant to 10 CFR 50.73(a)(2)(iv)(A), any event or condition that resulted in manual or automatic actuation of the Reactor Protection System.

EVENT CAUSE

The cause of the turbine trip signal was due to a Generator Side Disconnect [DISC] switch failing as a result of not being fully seated in the saddle of the disconnect fingers. This created a hotspot that melted the pads to the point where the cable connections faulted between the disconnect and the triad jumper cable to the next length of bus work. The damaged disconnects resulted in an open circuit condition on the output of the Main Generator.

After further investigation, it was determined that the disconnects were not properly seated in the saddle of the disconnect fingers because of personnel error. During the closing of the disconnects following maintenance activities, the disconnects were locked in a partially closed position. The movable members were closed, but not all the way to their mechanical stops.

SAFETY ASSESSMENT

There were no actual Nuclear or Radiological safety consequences due to this event. Thus, this event was of minimal significance to the health and safety of the public. The plant response to this event was bounded by the Updated Safety Analysis Report analysis of a Main Turbine Trip with Steam Bypass Valves/Pressure Regulators in service.



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	☐ 052		YEAR 2023	SEQUENTIAL NUMBER 006	REV NO. 00

NARRATIVE

CORRECTIVE ACTIONS

- Establish the expectation for the RBS Switchyard Coordinator or a Subject Matter Expert to be engaged for reference or oversight on all work being performed at the switchyard by RBS personnel.
- Communicate with the Operations Department that only qualified electrical switchyard workers will be allowed to operate disconnects at the switchyard going forward.
- Procedure revisions are planned to track the following (tracked by Corrective Action Process):
 - Add guidance to ensure only qualified switchyard workers operate disconnects.
 - Add guidance to ensure thermography is performed after closing disconnects to verify the disconnects are functioning as designed.

PREVIOUS SIMILAR OCCURRENCES

None

Energy Industry Identification System (EIIS) codes are identified in the text as [XX]. River Bend equipment codes are identified as (XX).